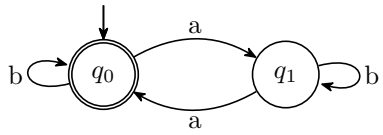
# オートマトン・言語理論 レポート課題 01

C5TB2006 安倍大陽

2026 年 10 月 21 日

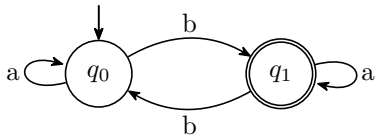
## 1 $L_1 = \{w \in \Sigma^* \mid \langle w \rangle_a = 2k, k \in \mathbb{N}\}$

$M = (\{q_0, q_1\}, \Sigma, \delta, q_0, \{q_0\})$



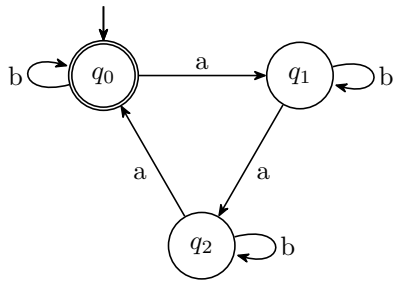

## 2 $L_2 = \{w \in \Sigma^* \mid \langle w \rangle_b = 2k + 1, k \in \mathbb{N}\}$

$M = (\{q_0, q_1\}, \Sigma, \delta, q_0, \{q_1\})$



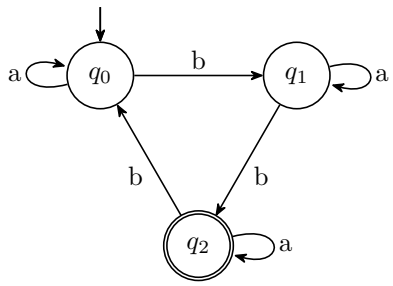

## 3 $L_3 = \{w \in \Sigma^* \mid \langle w \rangle_a = 3k, k \in \mathbb{N}\}$

$M = (\{q_0, q_1, q_2\}, \Sigma, \delta, q_0, \{q_0\})$



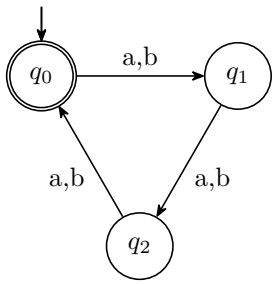
**4** $L_4 = \{w \in \Sigma^* \mid \langle w \rangle_b = 3k + 2, k \in \mathbb{N}\}$

$M = (\{q_0, q_1, q_2\}, \Sigma, \delta, q_0, \{q_2\})$



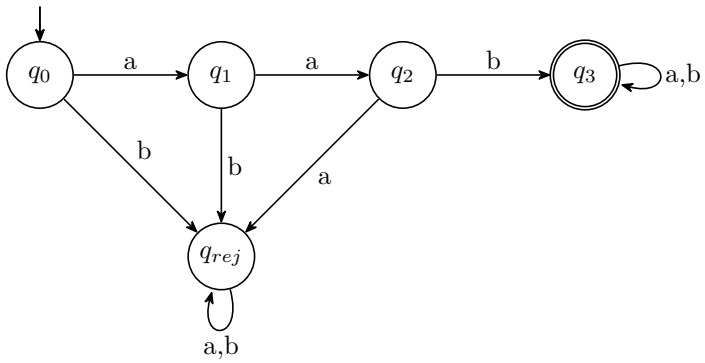

**5** $L_5 = \{w \in \Sigma^* \mid |w| = 3k, k \in \mathbb{N}\}$

$M = (\{q_0, q_1, q_2\}, \Sigma, \delta, q_0, \{q_0\})$



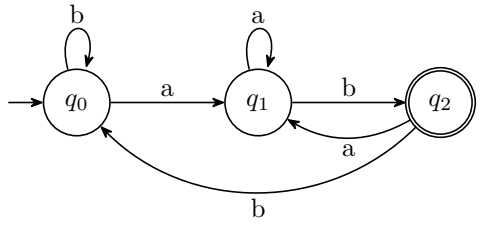

**6** $L_6 = \{\mathsf{aab}v \mid v \in \Sigma^*\}$

$M = (\{q_0, q_1, q_2, q_3, q_{rej}\}, \Sigma, \delta, q_0, \{q_3\})$



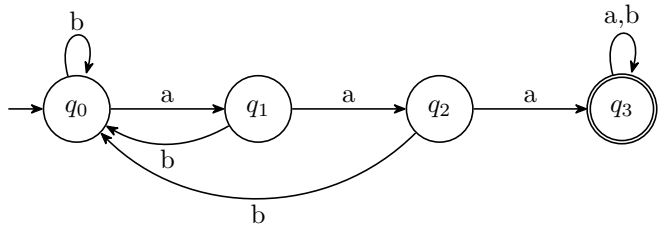
## 7 $L_7 = \{u\mathsf{ab} \mid u \in \Sigma^*\}$

$M = (\{q_0, q_1, q_2\}, \Sigma, \delta, q_0, \{q_2\})$



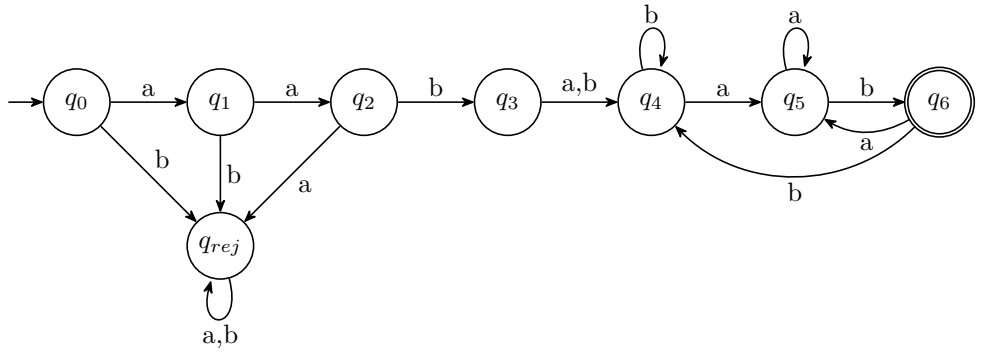

## 8 $L_8 = \{u\mathsf{aaa}v \mid u, v \in \Sigma^*\}$

$M = (\{q_0, q_1, q_2, q_3\}, \Sigma, \delta, q_0, \{q_3\})$



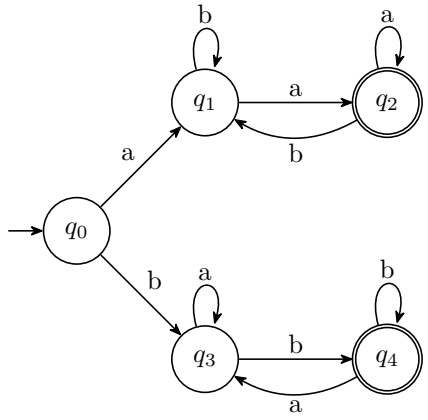

## 9 $L_9 = \{\mathsf{aab}v\mathsf{ab} \mid v \in \Sigma^*\}$

$M = (\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_{rej}\}, \Sigma, \delta, q_0, \{q_6\})$



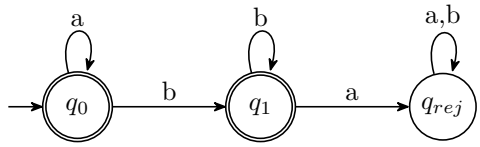
**10** $L_{10} = \{\mathsf{a}v\mathsf{a} \mid v \in \Sigma^*\} \cup \{\mathsf{b}v\mathsf{b} \mid v \in \Sigma^*\}$

$M = (\{q_0, q_1, q_2, q_3, q_4\}, \Sigma, \delta, q_0, \{q_4\})$



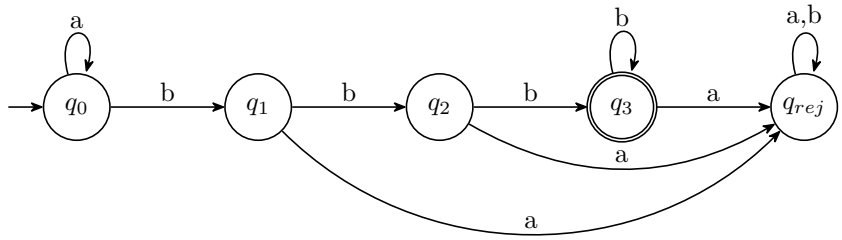

**11** $L_{11} = \{\mathsf{a}^i\mathsf{b}^j \mid i \geq 0, j \geq 0\}$

$M = (\{q_0, q_1, q_{rej}\}, \Sigma, \delta, q_0, \{q_0, q_1\})$

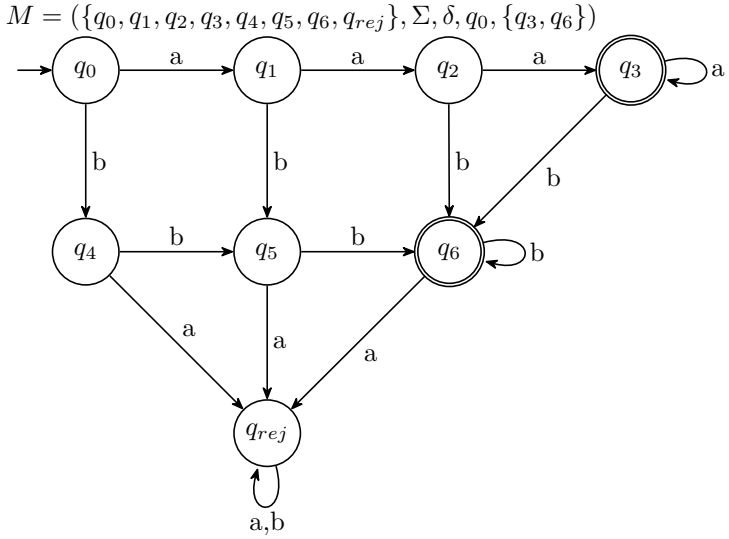



**12** $L_{12} = \{\mathsf{a}^i\mathsf{b}^j \mid i \geq 0, j \geq 3\}$

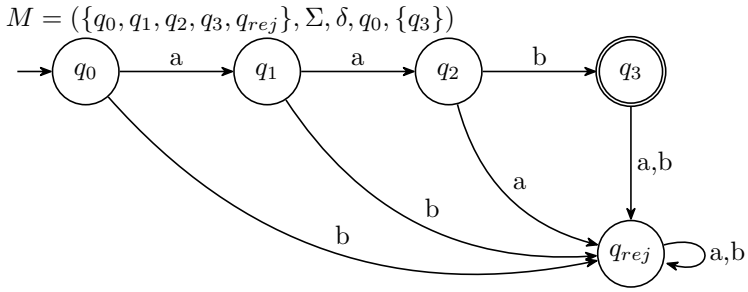
$M = (\{q_0, q_1, q_2, q_3, q_{rej}\}, \Sigma, \delta, q_0, \{q_3\})$



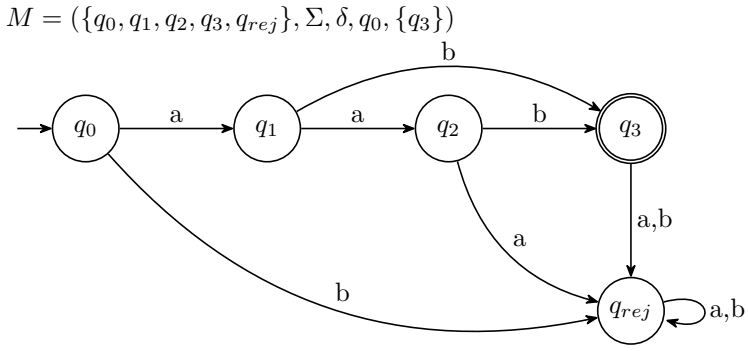
## 13 $L_{13} = \{\mathsf{a}^i\mathsf{b}^j \mid i + j \geq 3\}$




## 14 $L_{14} = \{\mathsf{aab}\}$

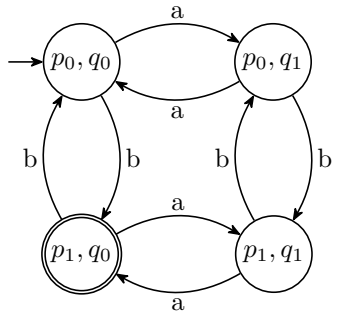



## 15 $L_{15} = \{\mathsf{aab}, \mathsf{ab}\}$



## 16 $L_{16} = L_1 \cap L_2$

$M = (\{p_0, p_1\} \times \{q_0, q_1\}, \Sigma, \delta, q_0, \{(p_1, q_0)\})$




## 17 $L_{17} = L_1 \cup L_2$

$M = (\{p_0, p_1\} \times \{q_0, q_1\}, \Sigma, \delta, q_0, \{(p_0, q_0), (p_1, q_0), (p_1, q_1)\})$

